CS383 · SOFTWARE ENGINEERING

What Repeats

An analysis of four midterms

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What's inside

01 The bottom line

02 The repetition table

03 What repeated word for word

04 How to answer each type

05 Study plan

01 The bottom line

I compared four midterms: 461 and three undated ones that I named Model 1, 2 and 3. The midterm is built from about five question types that stay the same while the numbers and texts change.

The three most important questions

Choosing the process model appeared in all four. Ethics appeared in three. FR/NFR appeared in three. These three carry 8 to 16 marks in every midterm.

02 The repetition table

The number in each cell is the question's mark.

Question type	Model 1	Model 2	Model 3	461	Lecture
Choosing the process model	4	4	4	3	3 · 4
Ethics: the eight IEEE principles	—	8	4	2	2
FR and NFR: extract or classify	4	4	—	5	7
MCQ	2	2	9	3	1 – 6
The four activities in order	—	2	—	2	3
Product / Organization / External	6	—	—	—	7
Sprint calculations	4	—	—	—	4
True / False with a reason	—	—	3	—	1 · 6
Total	**20**	**20**	**20**	**15**	

03 What repeated word for word

Question	Where	Answer
What does product quality depend on?	Model 2 · 461	Development process
...how easy it is to understand the process definition?	Model 2 · 461	Understandability
What are the 4 key software engineering activities... in a logical order	Model 2 · 461	Specification → Development → Verification and validation → Evolution
According to the 8 principles of the IEEE Code of Ethics...	Model 2 · 461	The same wording, with different statements.

The same ideas also returned in other forms: the product quality factors (Model 3), "NOT part of the process activity" (Model 3), and plan-driven versus agile (a table in 461, projects in the others).

04 How to answer each type

1 Choosing the process model

Signal in the text	Answer
Customer available daily, frequent change, a small co-located team.	**Agile**
Stable requirements, regulation, documentation and sign-offs, a large or distributed team, safety-critical.	**Plan-Driven**
COTS, an SDK or ready components that must be integrated.	**Component-Based**
Everything moderate, and "regular iterations".	**Iterative**

2 Ethics

Learn one keyword for each principle. The statements come from the text of the Code itself.

Principle	Keywords
PUBLIC	responsibility for their own work · safe for use
CLIENT AND EMPLOYER	confidential information
PRODUCT	appropriate method · integrity of data · achievable goals
JUDGMENT	conflicts of interest
MANAGEMENT	informed of standards · realistic estimation
PROFESSION	reputation of the profession
COLLEAGUES	colleague · fair hearing · assist colleagues
SELF	improve their knowledge or ability · prejudices

3 FR / NFR

- FR: an action the system or the user performs (create, update, search, record).
- NFR: a number or a quality (within 5 seconds, 100 concurrent users, encrypted, available).
- Copy the sentence from the text as it is. The student who got full marks did exactly that.

05 Study plan

- Start with Lecture 2 (ethics), Lecture 4 (choosing the model and sprint calculations) and Lecture 7 (requirements). They carry most of the marks.
- Memorise the four activities in order, the four product quality factors, and the four people-management factors.
- Solve Model 1 and Model 2 against a one-hour timer, then the mock.

Four midterms are a small sample. They show the style of the questions and do not guarantee what the next midterm will look like.